

Solutions to Problem Sheet II

Mathematics II (MAL1020)

Department of Mathematics, IIT Jodhpur

Problem 1

We are given the vectors

$$v_1 = (1, 2, 1, -1), \quad v_2 = (2, 1, -1, 1), \quad v_3 = (3, 3, 0, 0), \quad v_4 = (1, 5, 3, -3)$$

in $\mathbb{R}^4$.

(a) Expressing v_4 as a linear combination

We check whether there exist scalars $\alpha, \beta, \gamma \in \mathbb{R}$ such that

$$\alpha v_1 + \beta v_2 + \gamma v_3 = v_4.$$

This gives the system

$$\begin{aligned}\alpha + 2\beta + 3\gamma &= 1, \\ 2\alpha + \beta + 3\gamma &= 5, \\ \alpha - \beta &= 3, \\ -\alpha + \beta &= -3.\end{aligned}$$

The last two equations are equivalent and give

$$\alpha = \beta + 3.$$

(b) Solving for the scalars

Substituting $\alpha = \beta + 3$ into the first two equations:

$$(\beta + 3) + 2\beta + 3\gamma = 1 \Rightarrow 3\beta + 3\gamma = -2,$$

$$2(\beta + 3) + \beta + 3\gamma = 5 \Rightarrow 3\beta + 3\gamma = -1.$$

Since these two equations contradict each other, the system is inconsistent. Hence,

$$v_4 \notin \text{span}\{v_1, v_2, v_3\}.$$

(c) Uniqueness

Since v_4 cannot be expressed as a linear combination of v_1, v_2, v_3 , the question of uniqueness does not arise.

(d) Linear combinations yielding the zero vector

We solve

$$\alpha v_1 + \beta v_2 + \gamma v_3 = 0.$$

This leads to

$$\alpha + 2\beta + 3\gamma = 0,$$

$$2\alpha + \beta + 3\gamma = 0,$$

$$\alpha - \beta = 0,$$

$$-\alpha + \beta = 0.$$

From the last two equations, $\alpha = \beta$. Substituting into the first equation:

$$3\alpha + 3\gamma = 0 \Rightarrow \alpha = -\gamma.$$

Thus, all solutions are of the form

$$(\alpha, \beta, \gamma) = (-t, -t, t), \quad t \in \mathbb{R}.$$

—

Problem 2

Let

$$S = \{(1, 1, 1, 0), (2, 1, 0, 1), (3, 2, 1, 1), (1, 0, -1, 1)\} \subset \mathbb{R}^4.$$

(a) Linear independence

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 1 \\ 1 & 0 & -1 & 1 \end{bmatrix}$$

after calculating the row reduced echelon form we get

$$A' = \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

As we can see that two rows are zero so it is not linearly independent.

(b) Basis for $\text{span}(S)$

Removing the dependent vector, a basis is

$$\{(1, 1, 1, 0), (2, 1, 0, 1)\}.$$

(c) Dimension

Since the basis contains two vectors,

$$\dim(\text{span}(S)) = 2.$$

(d) Expressing a vector

We express $(4, 2, 0, 2)$ as a linear combination of the basis:

$$a(1, 1, 1, 0) + b(2, 1, 0, 1) = (4, 2, 0, 2).$$

Solving the corresponding system gives $a = 0, b = 2$.

Problem 3

Let

$$U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}, \quad W = \text{span}\{(1, 1, 1)\}.$$

(a) Direct sum decomposition

We need to prove that $\mathbb{R}^3 = U + W$ and $U \cap W = \phi$.

We can see by (c) that $U \cap W = \phi$.

For any $v = (a, b, c) \in \mathbb{R}^3$, define

$$w = \frac{a + b + c}{3}(1, 1, 1), \quad u = v - w.$$

Then $u \in U$ and $w \in W$, and $v = u + w$. Hence,

$$\mathbb{R}^3 = U \oplus W.$$

(b) Decomposition of a vector

For $v = (2, -1, 3)$,

$$w = \frac{4}{3}(1, 1, 1), \quad u = (2, -1, 3) - w.$$

(c) Intersection

If $(x, y, z) \in U \cap W$, then $(x, y, z) = t(1, 1, 1)$ and $3t = 0$, hence $t = 0$.

Problem 4

Let

$$U = \text{Col} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} = \text{span}\{(1, 1, 0), (0, 1, 1)\} \subset \mathbb{R}^3.$$

(a) Finding a subspace W

Any vector in U has the form

$$a(1, 1, 0) + b(0, 1, 1) = (a, a + b, b), \quad a, b \in \mathbb{R}.$$

Choose the vector $(1, -1, 1)$. Suppose it belongs to U . Then

$$(a, a + b, b) = (1, -1, 1).$$

This implies $a = 1$ and $b = 1$, but then $a + b = 2 \neq -1$, a contradiction. Hence $(1, -1, 1) \notin U$. Define

$$W = \text{span}\{(1, -1, 1)\}.$$

(b) Verification

Let $x \in U \cap W$. Then

$$x = (a, a + b, b) = t(1, -1, 1)$$

for some $a, b, t \in \mathbb{R}$.

Comparing components gives

$$a = t, \quad b = t, \quad a + b = -t.$$

Substituting yields $2t = -t$, hence $t = 0$. Therefore $x = (0, 0, 0)$ and

$$U \cap W = \{0\}.$$

Since $\dim(U) = 2$, $\dim(W) = 1$, and $U \cap W = \{0\}$, we conclude

$$\mathbb{R}^3 = U \oplus W.$$

(c) Decomposition of $(3, 2, 1)$

We seek

$$(3, 2, 1) = a(1, 1, 0) + b(0, 1, 1) + t(1, -1, 1).$$

This gives

$$(a + t, a + b - t, b + t) = (3, 2, 1).$$

Solving the system

$$\begin{cases} a + t = 3, \\ a + b - t = 2, \\ b + t = 1 \end{cases}$$

yields $t = 2/3$, $a = 7/3$, and $b = 1/3$.

Thus,

$$(3, 2, 1) = \frac{7}{3}(1, 1, 0) + \frac{1}{3}(0, 1, 1) + \frac{2}{3}(1, -1, 1),$$

.

Problem 5

Let

$$B = \{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$$

be an ordered basis of $\mathbb{R}^3$.

(a) Change-of-coordinates matrix $P_{B \rightarrow E}$

The change-of-coordinates matrix from basis B to the standard basis is formed by placing the basis vectors as columns:

$$P_{B \rightarrow E} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

This matrix satisfies

$$[v]_E = P_{B \rightarrow E}[v]_B \quad \text{for all } v \in \mathbb{R}^3.$$

(b) Inverse matrix $P_{E \rightarrow B}$

To find $P_{E \rightarrow B}$, we compute the inverse of $P_{B \rightarrow E}$ using row-reduction. Start with the augmented matrix:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right].$$

Row-reducing this matrix gives:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{array} \right].$$

Hence,

$$P_{E \rightarrow B} = \frac{1}{2} \begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}.$$

(c) Coordinates of $v = (2, 3, 1)$ relative to B

We compute

$$[v]_B = P_{E \rightarrow B}[v]_E = \frac{1}{2} \begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}.$$

This gives

$$[v]_B = \frac{1}{2} \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}.$$

Thus,

$$\boxed{[v]_B = (2, 1, 0)}.$$

—

Problem 6

(i)

Since $(1, -1)$ and $(2, 3)$ are linearly independent, the set spans $\mathbb{R}^2$.

(ii)

All vectors are scalar multiples of $(3, -1)$, so they do not span $\mathbb{R}^2$.

—

6. Determine whether the following sets of vectors span $\mathbb{R}^2$:

(i) $\{(1, -1), (2, -2), (2, 3)\}$.

Since $(2, -2) = 2(1, -1)$, it is enough to check $(1, -1)$ and $(2, 3)$:

$$\det \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} = 1 \cdot 3 - (-1) \cdot 2 = 5 \neq 0.$$

Hence $(1, -1)$ and $(2, 3)$ are linearly independent, so the set spans $\mathbb{R}^2$.

(ii) $\{(6, -2), (-2, 2/3), (3, -1)\}$.

$$(6, -2) = 2(3, -1), \quad (-2, 2/3) = \left(-\frac{2}{3}\right)(3, -1).$$

All vectors are scalar multiples of $(3, -1)$, so the span is 1-dimensional and does not equal $\mathbb{R}^2$.

7. Show that $\{(1, 2, 3), (3, 4, 5), (4, 5, 6)\}$ does not span $\mathbb{R}^3$, but spans a plane in $\mathbb{R}^3$ (find the plane equation).

Let

$$v_1 = (1, 2, 3), \quad v_2 = (3, 4, 5), \quad v_3 = (4, 5, 6).$$

Then

$$v_2 - v_1 = (2, 2, 2) = 2(1, 1, 1), \quad v_3 - v_2 = (1, 1, 1).$$

Hence $v_1, v_2, v_3 \in \text{span}\{v_1, (1, 1, 1)\}$, so $\dim(\text{span}\{v_1, v_2, v_3\}) \leq 2$. Therefore, the set does not span $\mathbb{R}^3$.

A normal vector to the plane is

$$n = v_1 \times (1, 1, 1) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{vmatrix} = (-1, 2, -1).$$

Thus the plane (through the origin) is

$$n \cdot (x, y, z) = 0 \iff -x + 2y - z = 0 \iff x - 2y + z = 0.$$

Hence the vectors span the plane $x - 2y + z = 0$.

8. Solve the following:

(i) In $\mathcal{M}_2(\mathbb{R})$, let

$$A_1 = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 1 \\ -2 & 1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}.$$

Let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. From $X = \lambda_1 A_1 + \lambda_2 A_2 + \lambda_3 A_3$, we get

$$a = \lambda_1 + 3\lambda_3, \quad b = -\lambda_1 + \lambda_2, \quad c = 2\lambda_1 - 2\lambda_2 + \lambda_3, \quad d = \lambda_2 + 2\lambda_3.$$

Eliminating $\lambda_1, \lambda_2, \lambda_3$ gives the single condition

$$-a + b + c + d = 0 \iff a = b + c + d.$$

Hence

$$\text{span}\{A_1, A_2, A_3\} = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathcal{M}_2(\mathbb{R}) : a = b + c + d \right\} = \left\{ \begin{pmatrix} b+c+d & b \\ c & d \end{pmatrix} : b, c, d \in \mathbb{R} \right\}.$$

(ii) In $\mathcal{M}_2(\mathbb{R})$, let

$$A_1 = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}, \quad A_2 = \begin{pmatrix} -2 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then

$$\text{span}\{A_1, A_2\} = \{\alpha A_1 + \beta A_2 : \alpha, \beta \in \mathbb{R}\} = \left\{ \begin{pmatrix} \alpha - 2\beta & 2\alpha + \beta \\ -\alpha + \beta & 3\alpha - \beta \end{pmatrix} : \alpha, \beta \in \mathbb{R} \right\}.$$

For $B = \begin{pmatrix} 3 & 1 \\ -2 & 4 \end{pmatrix}$, solve $\alpha A_1 + \beta A_2 = B$:

$$\alpha = 1, \quad \beta = -1.$$

Thus $B \in \text{span}\{A_1, A_2\}$, and in fact $B = A_1 - A_2$.

9. Determine whether the following vectors are linearly dependent/independent:

(i) In $\mathbb{R}^4$,

$$v_1 = (1, 4, 1, 7), \quad v_2 = (3, -5, 2, 3), \quad v_3 = (2, -1, 6, 9), \quad v_4 = (-2, 3, 1, 6).$$

Let $M = [v_1 \ v_2 \ v_3 \ v_4]$. Then

$$\det(M) = -462 \neq 0.$$

Hence $\{v_1, v_2, v_3, v_4\}$ is linearly independent.

(ii) In $P_1(\mathbb{R})$, $p_1(x) = a + bx$ and $p_2(x) = c + dx$. They are linearly dependent iff there exists λ such that $(a, b) = \lambda(c, d)$, i.e.

$$ad - bc = 0.$$

They are linearly independent iff $ad - bc \neq 0$.

10. Check whether $\{2 + x^2, 4 - 2x + 3x^2, 1 + x\}$ spans $P_2(\mathbb{R})$.

Write in the basis $\{1, x, x^2\}$:

$$2 + x^2 = (2, 0, 1), \quad 4 - 2x + 3x^2 = (4, -2, 3), \quad 1 + x = (1, 1, 0).$$

The determinant of the coefficient matrix is 0 (rank = 2), so the set is linearly dependent and does not span $P_2(\mathbb{R})$. Also,

$$-3(2 + x^2) + (4 - 2x + 3x^2) + 2(1 + x) = 0 \iff 4 - 2x + 3x^2 = 3(2 + x^2) - 2(1 + x).$$

Hence $\text{span}\{2 + x^2, 4 - 2x + 3x^2, 1 + x\} = \text{span}\{2 + x^2, 1 + x\}$.

11. Let $p_1(x) = 1 + x$, $p_2(x) = x(x - 1) = x^2 - x$, $p_3(x) = 1 + 2x^2$. For $p(x) = a_0 + a_1x + a_2x^2$, find coordinates of p w.r.t. $\{p_1, p_2, p_3\}$.

Write

$$p = \alpha p_1 + \beta p_2 + \gamma p_3.$$

Use values at simple points.

At $x = 0$:

$$p(0) = a_0 = \alpha p_1(0) + \beta p_2(0) + \gamma p_3(0) = \alpha + 0 + \gamma,$$

so

$$\alpha + \gamma = a_0. \quad (1)$$

At $x = 1$:

$$p(1) = a_0 + a_1 + a_2 = \alpha p_1(1) + \beta p_2(1) + \gamma p_3(1) = 2\alpha + 0 + 3\gamma,$$

so

$$2\alpha + 3\gamma = a_0 + a_1 + a_2. \quad (2)$$

At $x = -1$:

$$p(-1) = a_0 - a_1 + a_2 = \alpha p_1(-1) + \beta p_2(-1) + \gamma p_3(-1) = 0 + 2\beta + 3\gamma,$$

so

$$2\beta + 3\gamma = a_0 - a_1 + a_2. \quad (3)$$

From (1), $\alpha = a_0 - \gamma$. Put into (2):

$$2(a_0 - \gamma) + 3\gamma = a_0 + a_1 + a_2 \Rightarrow 2a_0 + \gamma = a_0 + a_1 + a_2 \Rightarrow \gamma = -a_0 + a_1 + a_2.$$

Then

$$\alpha = a_0 - \gamma = 2a_0 - a_1 - a_2.$$

Now use (3) to find β :

$$\begin{aligned} 2\beta + 3(-a_0 + a_1 + a_2) &= a_0 - a_1 + a_2 \\ 2\beta &= a_0 - a_1 + a_2 + 3a_0 - 3a_1 - 3a_2 = 4a_0 - 4a_1 - 2a_2 \\ \beta &= 2a_0 - 2a_1 - a_2. \end{aligned}$$

Therefore,

$$[p]_{\{p_1, p_2, p_3\}} = (\alpha, \beta, \gamma) = (2a_0 - a_1 - a_2, 2a_0 - 2a_1 - a_2, -a_0 + a_1 + a_2).$$